5.3.4

AI25BTECH11010 - Dhanush Kumar

Question

Solve for $x \in \mathbb{N}$ using matrices: $\det\begin{pmatrix} x+3 & -2 \\ -3x & 2x \end{pmatrix} = 8$

Solution

$$\det\begin{pmatrix} x+3 & -2 \\ -3x & 2x \end{pmatrix} = (x+3)(2x) - (-2)(-3x) \tag{1}$$

$$= 2x^2 + 6x - 6x \tag{2}$$

$$= 2x^2 \tag{3}$$

$$2x^2 = 8 \tag{4}$$

$$x^2 = 4 \tag{5}$$

$$x = \pm 2 \tag{6}$$

$$x \in \mathbb{N} \implies x = 2 \tag{7}$$